Muhammet Celal Gokalp

Application Engineer | Hands-on Vehicle & Hardware Systems

+44 07397791625 | muhammetcelalgokalp@gmail.com | Coventry, CV5 8HN, UK | British, Turkish
linkedin.com/in/celal-gökalp-4616ab122

PROFESSIONAL SUMMARY

Mechanical engineer with nearly 8 years in automotive engineering, including 2 years based at a customer site (Jaguar Land Rover). Comfortable with hands-on work: testing in vehicles and on test rigs, finding faults on the vehicle network, and working with suppliers to fix them. Outside work, I built a 6-DOF motion platform at university and I repair my own cars at home. I enjoy explaining technical topics simply, from teaching science at Space Camp Turkey to supporting engineers at customers. I learn new systems quickly and I am happy to travel for on-site work.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch — Cambridge, United Kingdom

- Technical link between the core engineering teams and enterprise customers using our simulation tools for automated driving.
- On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member; triage and fix toolchain and map integration issues (SDB/XODR) so customers can keep working without interruption.
- Build small internal tools (e.g. a GUI for JSON parameter set-up and a local STAX uploader) that make day-to-day work faster for the wider team.
- Write and run test scenarios (OpenSCENARIO 2, Euro NCAP, UN R79 lane change) and review the results to find and report issues.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- Based at the Jaguar Land Rover site for 2 years as AVL's technical contact for ADAS features (Rear Cross Traffic Alert, Blind Spot Monitor, Rear Pre-Crash).
- Tested features in vehicles on the proving ground and on Hardware-in-the-Loop (HiL) and Vehicle-in-the-Loop (ViL) rigs.
- Found faults by connecting to the vehicle CAN network with Vector CANoe and CANape and reading the data logs.
- At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- Worked with suppliers and engineers to fix defects, e.g. adjusting a sensor's field of view when an alert triggered late, and tracked fixes in Rational Team Concert.
- Wrote requirements and test cases in DOORS and MKS Integrity within an ASPICE V-model process.

Mechanical Engineer Intern

05/2014 – 08/2014

Temsa Global (bus manufacturer) — Adana, Türkiye

- Worked on quality improvements on the bus production line.
- Calculated power and torque needs for a new electric bus powertrain in MATLAB/Simulink.

Part-Time Teacher and Program Planner

12/2013 – 01/2017

Space Camp Turkey (associated with NASA) — Izmir, Türkiye

- Taught space technology, physics, chemistry, and biology to students aged 9 to 16, using simple, hands-on explanations.
- Planned daily schedules for student groups and teachers.

HANDS-ON PROJECTS

6-DOF Stewart Motion Platform (BSc dissertation)

2016 – 2017

Ege University

Izmir, Türkiye

- Designed the platform in SolidWorks, built the physical prototype, modelled its movement in MATLAB, and wrote the control code on Arduino.

Home Car Repairs

Ongoing

Personal

Coventry, United Kingdom

- Buy, diagnose, and repair my own cars at home (including a Peugeot 306 HDi and a Ford Focus), using workshop manuals and wiring diagrams.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation on the thermal model of an electric motor stator, including lab testing; MATLAB, Simulink, MotorCAD.

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: 6-DOF Stewart motion platform (see Hands-on Projects).

SKILLS

Hands-on & Diagnostics	Fault-finding, vehicle testing, HiL/ViL test rigs, Vector CANoe / CANape, prototype build, home mechanical repairs
Customer & Communication	On-site customer work (2 years at JLR), supplier coordination, teaching and explaining technical topics
Software & CAD	Python, MATLAB / Simulink, SolidWorks, Git, Jira, Confluence
Languages	English (C2), Turkish (Native)
Other	Full UK driving licence; happy to travel